Huronian polar wander and paleomagnetism of the Thessalon volcanics

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The Thessalon basalts outcrop in Ontario on the north shore of Lake Huron near the base of the Huronian Supergroup in the Southern tectonic province. After alternating field and thermal demagnetization on 230 specimens from 23 sites, a stable high coercivity remanence in titanomagnetite was isolated giving a direction of 304°, 54° (α_{95} − 7°). The pole position of 169°W, 47°N (7°, 9°) corresponds to the primary age of 2375 ± 75 Ma based on geologic evidence and radiometric data from other units. Thus, the magnetite remanence predates the reset Rb–Sr isochron age of 2040 ± 50 Ma determined for the basalts. Viscous remanence directions confirm the petrologic evidence that the basalts have not been metamorphosed much above lowest greenschist facies. Existing North American Paleoaphebian paleomagnetic data are critically evaluated and a conservative apparent polar wander path drawn. The Thessalon volcanics were extruded at a middle paleolatitude of 35° indicating that the contemporaneous Blind River – Elliot Lake uranium deposits were formed in a warm temperate climate.

Les basaltes de Thessalon affleurent en Ontario sur la rive nord du lac Huron près de la base du supergroupe de l'Huronien dans la province tectonique du Sud. Après désaimantation thermique et dans un champ alternatif de 230 spécimens provenant de 23 sites, on a isolé une rémanence stable de coercivité élevée dans la titanomagnétite avec une direction de 304°, 54° (α_{95} −7°). La position du pôle à 169°O, 47°N (7°, 9°) correspond à l'âge primaire de 2375 ± 75 Ma basé sur les données géologiques et radiométriques provenant d'autres unités. Ainsi, la rémanence de la magnétite précède l'âge de réouverture de l'isochrone Rb–Sr de 2040 ± 50 Ma déterminé pour les basaltes. Les directions de rémanence visqueuse confirment les données pétrologiques à l'effet que les basaltes n'ont pas été métamorphisés beaucoup au-dessous du faciès des schistes verts. On évalue de façon critique les données actuelles sur le paléomagnétisme du Paléoaphébien d'Amérique du Nord et on trace une trajectoire probable de dérive polaire apparente. Les roches volcaniques de Thessalon ont été mises en place à une paléolatitude moyenne de 35°, indiquant ainsi que les dépôts contemporains d'uranium de Blind River – Elliot Lake se sont formés dans un climat tempéré chaud.

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Introduction

The Thessalon volcanics outcrop on the north shore of Lake Huron near the town of Thessalon, Ontario (Fig. 1). They are accessible from Highways 17 and 129 and from a number of intersecting concession roads (Fig. 2).

The Thessalon volcanics are mainly mafic, unaltered in appearance, and gently deformed. They occur close to the base of the thick Huronian Supergroup in the Penokean fold belt of the Southern tectonic province (Fig. 1). The volcanics give Rb–Sr dates of between 1797 ± 24 and 2065 ± 40 Ma. Contrarily all other geologic and radiometric age dates limit the age to ~2375 ± 75 Ma. This study was undertaken: (1) to provide a primary pole position for the volcanics and thereby help resolve the conflict in dates; (2) to add to the North American Paleoaphebian polar wander path; and (3) to place paleoclimatic limits on the Huronian Supergroup with its 'tillites' and uranium deposits.



FIG. 1. Location map.

Geology

There are three significant outcrop areas of Thessalon basic volcanics in the Penokean fold belt of the Southern geologic province (Stockwell 1969). Two of the areas are unsuited to paleomagnetic study because the volcanics exhibit extensive chloritic alteration and deformation schistosity. These areas are the 20 × 4 km belt centred in Dun-

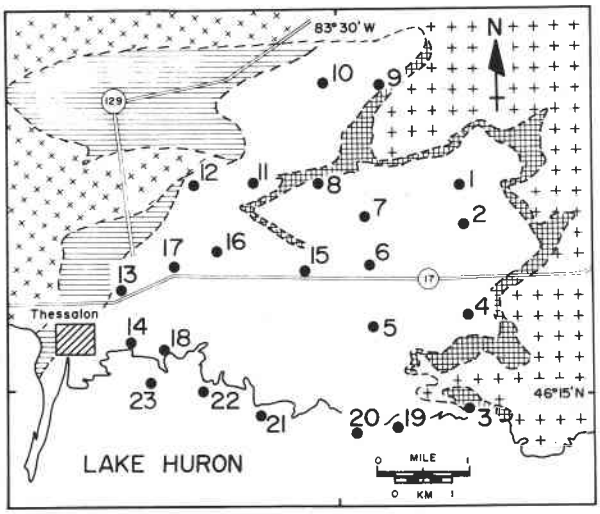


FIG. 2. Site location and general geology of the Thessalon outcrop area. The Archean granite is shown by crosses, the Livingstone Creek quartzite by squared pattern, the Thessalon volcanics by no pattern, the Mississagi quartzite by horizontal lines, and the Nipissing diabase by X pattern.

can Twp. some 15 km northeast of Sault Ste. Marie (McConnell 1927; Hay 1964; Frarey 1978) and the 11 × 3 km belt centred in Aberdeen Twp. some 35 km to the northwest of Thessalon, Ontario (Frarey 1959, 1962, 1978). The studied outcrop area is about 10 km in diameter. It is located immediately northeast of the town of Thessalon, Ontario (Fig. 2). Here the Thessalon Formation is comprised of dark grey to black, fine-grained, massive, subareal, tholeiitic basalt. Intercalated lenses of rhyolite, quartzite, argillite, and quartz-pebble conglomerate form mappable units over short distances. The flows are rarely porphyritic, flow banded, or pillowed. Occasional amygdaloidal layers indicate that dips are gentle (<20°) and variable. From drilling, the Thessalon volcanics are known to be ≈700 m thick. The volcanics conformably overly the Livingstone Creek feldspathic quarzites that contain a basal uraniferous quartz-pebble conglomerate. This unit is thought to be equivalent to the Matinenda Member found at the base of the Mississagi Formation in the Huronian Supergroup. The Livingstone Creek – Thessalon sequence rests unconformably on a stable Archean granitic basement. The volcanics are overlain by apparently conformable Mississagi feldspathic quartzites. In thin section the volcanics are fine-grained mosaics of andesine plagioclase, hornblende, titanomagnetite, and some biotite (Frarey 1961*a*,*b*, 1978). Burial metamorphism to the low greenschist level has led to the formation of some albite, chlorite, occasional stilpnomelane, and traces of very fine biotite (G. Bennett, personal communication, 1978). Church (W. R. Church, personal communication, 1978) reports the presence of epidote and actinolite. In four sections examined by the authors, all minerals except stilpnomelane and actinolite were observed, and confirm a very low greenschist grade of metamorphism. The volcanics are cut by Nipissing diabase dikes. Regional metamorphism of the Huronian sediments and Nipissing dikes close to the volcanics has similarly reached a maximum of the lowermost greenschist facies.

Radiometric Ages

Rb–Sr whole-rock ages have been determined for the Thessalon volcanics by Van Schmus (1965) who obtained values of 1915 ± 50, 1940 ± 115, and 2065 ± 40 Ma on rhyolite members. Knight and Fairbairn (1966) (see also Fairbairn *et al.* 1969) obtained Rb–Sr ages of 1797 ± 24 Ma for metamorphosed basalt from the Aberdeen Twp. belt and of 1959 ± 112 Ma for rhyolite from close to Thessalon. Their combined isochron is 2018 ± 49 Ma. Equivalent Huronian metasediments and basic metavolcanics from the Espanola–Sudbury area to the east give a Rb–Sr isochron age of 1950 ± 100 Ma (Fairbairn *et al.* 1969). Based on this data, the best estimate for the age of the Thessalon volcanics would be 2040 ± 50 Ma.

The Archean basement rocks beneath the Thessalon volcanics give 2450–2700 Ma ages using a variety of radiometric techniques (Roscoe 1973). The stratigraphically equivalent Copper Cliff rhyolites near Sudbury give two Rb–Sr whole-rock ages of 2350 Ma (Fairbairn 1965). Unmetamorphosed Gowganda Formation sediments from above the Thessalon volcanics in the upper part of the Huronian Supergroup give a well-defined Rb–Sr whole-rock isochron of 2288 ± 87 Ma (Fairbairn *et al.* 1969). In addition, the Huronian sequence has been intruded by Nipissing diabase yielding Rb–Sr whole-rock isochrons of 2155 ± 80 Ma (Van Schmus 1965) and 2162 ± 27 Ma (Fairbairn *et al.* 1969) as well as biotite ages of 2180 ± 40 Ma using Rb–Sr (Van Schmus 1975) and 2095 Ma using K–Ar (Lowdon *et al.* 1963). Based on this data, the best estimate for the age of the Thessalon volcanics would be 2375 ± 75 Ma. Van Schmus (1965), Knight and Fairbairn (1966), Fairbairn *et al.* (1966), and Frarey (M. J. Frarey, personal communication, 1975) accept the argument that the Rb–Sr system in the Thessalon volcanics was opened by subsequent metamorphism during the Penokean (~2150 Ma (Church 1968)) or Hudsonian (~1850 Ma (Stockwell 1973)) orogenies thereby yielding the younger ages.

Experimental Methods

At each of 23 sites distributed uniformly throughout the outcrop area (Fig. 2), five cores (115 total) were drilled and oriented *in situ* using a solar compass. Two specimens (230 total) were cut from each core and measured on a Schonstedt SSM-IA spinner in a shielded room to determine their natural remanent magnetization (NRM). One pilot specimen per site with an average NRM direction and intensity for the site was cleaned using alternating field (AF) demagnetization at steps of 4.0, 8.0, 11.9, 15.9, 19.9, 27.8, 39.8, 55.7, and 71.6 kA/m using a Schonstedt GSD-1 demagnetizer.[1] Plots of the decay in remanence intensity and of the paleomagnetic stability index (PSI) (Symons and Stupavsky 1974) were used to select the optimum AF for cleaning the remaining specimens from each site. Stable end points with PSI values <600 μdeg/(A/m) were isolated for all sites between 15.9 and 71.6 kA/m. Sites 1, 6, and 21 gave two PSI 'lows' and were cleaned at two field strengths. The site statistics after AF cleaning are given in Table 1. In addition, 18 specimens from different sites were thermally demagnetized using an air-cooled, non-magnetic, induction furnace in nested mu-metal shields at temperature steps of 100, 200, 300, 400, 500, 525, 550, 600, and 650°C.

AF Site Statistics

First, the traditional approach to analysing paleomagnetic data was used. Those cores in which the two specimen directions diverged by >25° after AF demagnetization were regarded as inhomogenously magnetized and their data were rejected. At sites 1, 6, and 21 the lower AF cleaning field produced the better and accepted grouping of directions. At sites 4 and 13, only one core was accepted. Their data were also rejected as insufficient to represent the site. The core mean directions were combined to obtain the site mean directions and statistics (Table 1). The 18 sites with a cone of 95% confidence (α_{95}) (Fisher 1953) of <25° were considered to be sufficiently well clustered to be reliable. Their means appeared to form one population (Fig. 3) and were combined to give the formation mean direction (Table 2).

AF Step Demagnetization

The AF step demagnetization data can be analysed more fully. A first approach is to calculate the mean intensity and direction of the remanence for the specimens from the 18 accepted sites after each cleaning treatment. Within narrow limits, all

[1]SI units: 1 Oe = 0.0796 kA/m; 1 emu/cc = 10^6 mA/m; and 1 mdeg/Oe = 12.6 μdeg/(A/m).

TABLE 1. Remanence directions by sites

Site	AF clean (kA/m)	Cores*	Mean remanence Decl. (deg)	Incl. (deg)	α_{95} (deg)
1	15.9	5, 5	298	64	11†
	71.6	5, 2	293	63	81
2	39.8	5, 4	330	62	14†
3	27.8	5, 3	286	60	15†
4	15.9	5, 1			
5	39.8	5, 4	307	46	21†
6	19.9	5, 4	329	29	24†
	71.6	5, 3	318	27	27
7	15.9	5, 4	303	53	8†
8	19.9	5, 5	317	53	9†
9	39.8	5, 4	285	41	25†
10	15.9	5, 5	304	60	11†
11	27.8	5, 3	285	44	29
12	11.9	5, 3	264	62	61
13	55.7	5, 1			
14	39.8	5, 5	304	57	7†
15	4.0	5, 2	317	70	25†
16	15.9	5, 3	331	41	43
17	39.8	5, 4	313	50	16†
18	4.0	5, 4	303	47	16†
19	55.7	5, 4	309	53	19†
20	39.8	5, 3	264	63	13†
21	27.8	5, 4	334	48	18†
	71.6	5, 2	336	57	38
22	19.9	5, 5	272	46	21†
23	27.8	5, 4	290	42	25†

*Cores: collected, accepted.
†Sites accepted for further analysis (Table 2).

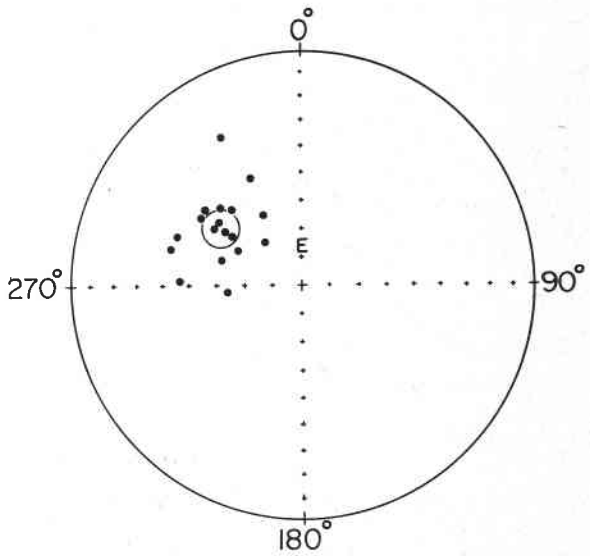


FIG. 3. Site mean remanence directions for the Thessalon volcanics. The directions are plotted on an equal area projection with their cone of 95% confidence. E is the Earth's present magnetic field direction.

TABLE 2. Grouped remanence directions

Group	Population		Mean remanence		
	Cores	Sites	Decl. (deg)	Incl. (deg)	α_{95} (deg)
Accepted sites ($\alpha_{95} < 25$)					
(a) From Table 1	73	18	304.0	53.9	6.6
(b) After tilt correction			304.1	51.7	6.8
AF step demagnetization 15.9 kA/m	18		306.9	49.9	9.9
AF component analysis using MDF > 79 kA/m	20		307.2	52.4	6.6
Thermal step demagnetization					
(a) At 100°C	18		306.4	53.1	11.7
(b) At 525°C	18		286.7	54.6	26.8

18 specimens of Thessalon volcanics behave in the same way. The mean direction shows a slight shallowing of inclination up to 27.8 kA/m indicating the removal of some Recent viscous remanence. This is followed by steepening of inclination up to 71.6 kA/m to return to the NRM direction (Fig. 4*a*). The scatter of directions about the mean increases slightly from $\alpha_{95} \sim 9°$ at 0 kA/m to ~13° at 39.8 kA/m, and then it increases sharply to ~20° at 71.6 kA/m. This increase is attributed to the addition of anhysteretic remanence (ARM) components to the stable components by the demagnetization process. The mean intensity of the remanence undergoes exponential decay from 0 to 39.8 kA/m as the stable component is progressively removed from each specimen (Fig. 4*b*). The decay rate changes above 39.8 kA/m reflecting the resultant of the exponential decay of the stable component and the exponential addition of the ARM component. Using this approach the mean direction at 15.9 kA/m cleaning reasonably represents the stable direction (Table 2).

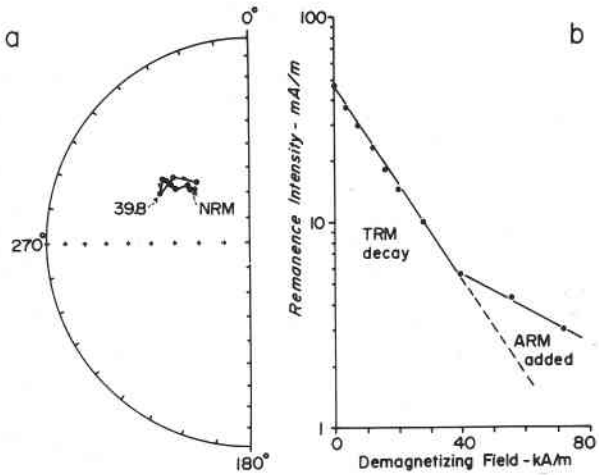


FIG. 4. AF demagnetization data. (*a*) The mean direction of all 18 specimens from accepted sites shown after cleaning at NRM or 0 kA/m (α_{95} – 9°), 4.0 (9°), 8.0 (9°), 11.9 (11°), 15.9 (10°), 19.9 (11°), 27.8 (13°), 39.8 (13°), 55.7 (20°), and 71.6 (19°) plotted on an equal-area projection. (*b*) The mean remanence intensity of the 18 specimens in mA/m on a log-linear plot to show exponential decay plotted against the demagnetizing field in kA/m.

The second approach to the AF step demagnetization data is to use the least-squares computer analysis method of Stupavsky and Symons (1976) to separate out the true stable high coercivity remanence component for each step demagnetized specimen. A high coercivity component with a median destructive field (MDF) of >80 kA/m was isolated for 20 sites. After 9999 interations, a stable direction was not isolated in the remaining three sites. Combining the 20 high coercivity component directions gives the same formation mean direction and scatter as determined using traditional techniques (Table 2, Fig. 5).

Thermal Step Demagnetization

The 18 Thessalon volcanic specimens subjected to thermal step demagnetization range in NRM intensity from 0.36 to 250 m/Am with a median of 2.3 mA/m. Nevertheless, all show a smooth linear decrease in intensity up to 500°C (Fig. 6*b*). Thus, 75% of their thermal blocking temperature spectrum is distributed between 0 and 500°C. From 500 to 600°C, the intensity shows an abrupt decrease as the Curie temperature of magnetite is approached and exceeded. Thus, 25% of their spectrum is discreetly confined to this temperature range. Above 600°C, the intensities increase rapidly indicating the production of remanence components by thermochemical reaction.

Between 0 and 200°C the mean thermal remanence direction decreases in inclination away from

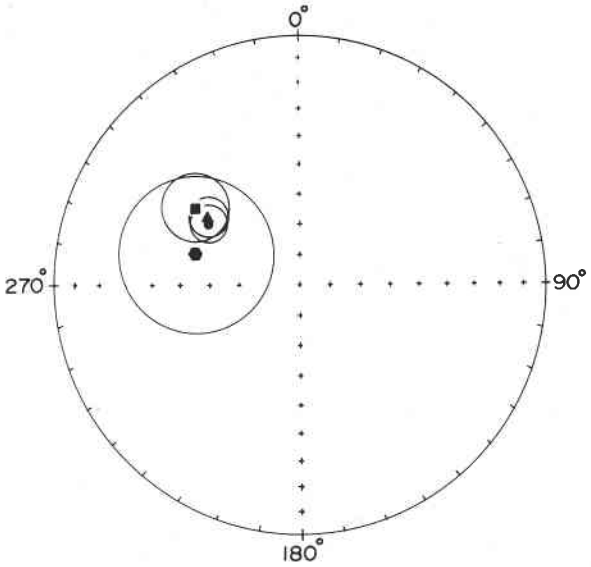


FIG. 5. The mean remanence direction for the Thessalon volcanics based on the 18 accepted site mean remanence directions (circle), on the 18 step demagnetized specimens after cleaning at 15.9 kA/m (square), on the 20 high coercivity (MDF > 80 kA/m) component directions from AF step demagnetization (triangle), and on the 18 specimens after thermal cleaning at 525°C (hexagon) as given in Table 2.

the present Earth's magnetic field direction as Recent viscous components are removed. Between 200 and 500°C, the direction is unchanged with the scatter increasing slightly with α_{95} going from 14 to 18° (Fig. 6*a*). At 525°C the direction remains unchanged but the scatter begins to increase significantly with an α_{95} of 26°. By 550°C the cluster is demolished with an α_{95} of >90°. A new cluster is not formed above 600°C. It is noteworthy, however, that four sites retain the stable direction to above 550°C before moving, and one site to above 600°C. The behaviour of the directions is consistent with the stable remanence of the Thessalon volcanics residing primarily in magnetite over a broad coercivity spectrum including a discrete high coercivity fraction.

Metamorphism

Based on the theoretical discussion of Pullaiah *et al.* (1975), the median destructive temperature (MDT) of the Thessalon volcanics is 305°C so that the stable component could have been produced by burial metamorphism at ~130°C for ~10^7 years. This does not appear to be a realistic approach because the Thessalon volcanics appear to have two domain populations in magnetite carrying the same stable direction. The distributed 0–500°C blocking temperature spectrum is attributed to multidomain magnetite grains and the discrete 500–600°C spectrum to single or pseudosingle domain magnetite. This observation was confirmed by the examination of four polished sections under magnifications up to ≈1300×. The sections showed numerous very fine-grained titanomagnetite crystals of about single domain size and a very few coarser euhedral crystals of titanomagnetite. The distributed population could have been progressively reset by metamorphism up to 400°C; however, the discrete population would only be progressively reset between 400 and 525°C. The greenschist facies reflects regional metamorphism temperatures in the order of 200–500°C (Blackwell 1971; Miyashiro 1972; Ernst 1973). From thin section observation, however, the Thessalon volcanics were never heated to more than ~250°C. Further, the titanomagnetite is all Class 1 type with the coarser crystals showing granulation. Ade-Hall *et al.* (1968) have shown that this assemblage is caused by burial metamorphism to the upper zeolite facies whereas higher metamorphism would cause exsolution of the titanomagnetite. Therefore, the stable direction isolated by traditional AF methods, AF step demagnetization analysis, and by thermal step demagnetization is the primary thermoremanent magnetization (TRM) (Fig. 5).

Fold Test

Frarey (1978) mapped a gentle anticlinal–synclinal structure in the volcanics at Thessalon based on the attitude of vesicular zones in the lavas and of adjacent strata, the shape of the volcanic outcrop area, and drill data. The limbs dip <15° northeast or southwest. Tilt corrections change neither the mean direction nor its α_{95} value significantly (Table 2) so that the test is inconclusive.

Age of Remanence

Using the 18 site mean directions the pole position for the Thessalon volcanics is 168.7°W, 46.8°N (6.5°, 9.2°).

Recently constructed apparent polar wander (APW) paths for the North American Precambrian Shield during the time interval from 2150 to 1750 Ma migrate through the present area of Central and South America and into the equatorial Pacific (e.g. Christie *et al.* 1975; Irving and Lapointe 1975; Symons 1975; Roy and Lapointe 1976; Cavanaugh and Seyfert 1977; Irving and Naldrett 1977; Morris 1977). The pole for the Thessalon volcanics is near the Aleutian Islands so that their remanence could not have been acquired during the Penokean or Hudsonian orogenies. The Thessalon pole is close to these APW paths during the 2200–2500 Ma interval, ≈1150 Ma ago, and

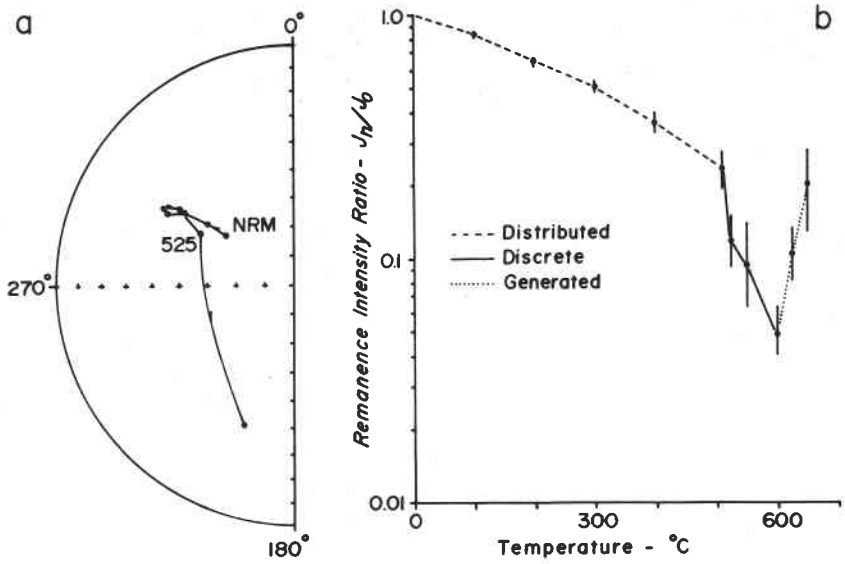


FIG. 6. Thermal demagnetization data. (*a*) The mean direction of the 18 specimens that were thermally demagnetized after cleaning at NRM or 0°C (α_{95} −11°), 100°C (12°), 200°C (14°), 300°C (15°), 400°C (16°), 500°C (18°), 525°C (27°), and 550°C (83°) plotted on an equal area net. (*b*) The mean of the ratio of the measured remanence intensity (J_n) to the NRM intensity (J_o) for the 18 specimens with the vertical standard error bars plotted against the demagnetizing temperature in °C to emphasize: (1) the 'distributed' and 'discrete' portions of the thermal coercivity spectrum and (2) the generation of significant thermochemical remanence components above 600°C.

≈100 Ma ago. There is no evidence for any mid-Neohelikian (≈1150 Ma) or Cretaceous (≈100 Ma) events in the Thessalon area. Thus, the first conclusion is that the volcanics acquired their remanence during the 2500–2200 Ma interval, and the stable high temperature remanence is a primary TRM acquired during extrusion ≈2375 ± 75 Ma ago.

The second conclusion is that the Thessalon volcanics at Thessalon have not been significantly metamorphosed to above upper zeolite–pumpellyite temperatures (150–250°C) during Penokean or later orogenies. This is evident because all secondary lower temperature components observed in the specimens have Huronian directions. It is possible that Huronian burial metamorphism could have set the secondary directions.

The third conclusion is that magnetite remanence can survive undisturbed even when the Rb–Sr radiometric age dating system is opened and reset. Presumably, the resetting of the Thessalon volcanic Rb–Sr (and K–Ar) age dates to <2150 Ma was induced primarily by chemical effects rather than by elevated temperatures. Thus paleomagnetic arguments based on the assumption that a magnetite pole correlates to an age younger than the Rb–Sr dates and about equal to the K–Ar dates may not be well founded (e.g. Irving *et al.* 1974; McWilliams and Dunlop 1975) as Buchan *et al.* (1977) have shown for some Grenville data.

Huronian Polar Wander

Construction of a Huronian APW path is difficult because most poles have limitations including insufficient sampling, assigning a magnetization age, and possible tectonic rotations. Huronian sedimentation occurred between 2450 and 2200 Ma so that only poles with assigned ages between 2600 and 2050 Ma are considered.

The Dogrib dikes (pole 1, Fig. 7) (McGlynn and Irving 1975) of the Slave tectonic province give radiometric ages from ~1800 to ~2700 Ma, cut rocks giving ages from ~2450 to ~2650 Ma, and have been assigned an age of ~2600 Ma. McGlynn and Irving (1975) and Cavanaugh and Seyfert (1977) invoke major plate tectonic movement between the Slave and the Superior–Southern provinces to explain this pole. The X dikes (pole 2; McGlynn and Irving 1975) have not been radiometrically dated but are thought to be the same age (~2600 Ma) as the Dogrib dikes. They also come from the Slave province, and only three sites were sampled to give the pole. The Thistlewaite dike, Duck Lake sill, Pensive Lake sheet (McGlynn and Irving 1975), and the unrecrystallized Beartooth Mountain dike (Larson *et al.* 1973) were sampled at one site only and so their poles are not shown.

Dunlop and Buchan (1976) believe that the Shelley Lake granite pole (pole 3) from the Superior province was set during Kenoran metamorphism ~2500 Ma although they do not give direct

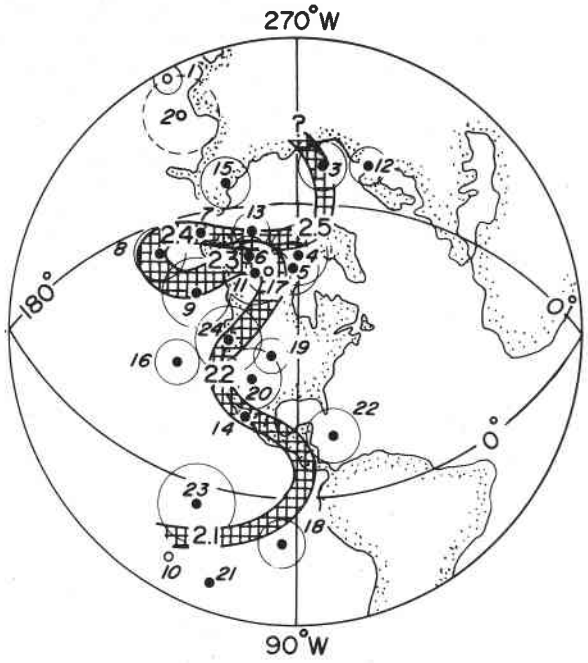


FIG. 7. Polar wander curve for the Huronian of North America. The individual poles are listed by number in the text and are shown with their cone of 95% confidence. The Earth projection is centred on 90°W, 45°N. The path (cross hatching) with time points in Ga is shown as a band ≈7° wide for clarity, however, it should have a width of ≈20° for 95% confidence based on well-defined paths such as the Cenozoic–Mesozoic path of North America and this would permit removal of some of the 'hairpins'. Poles from the discordant Slave province are shown as open circles including pole 2 that is over the horizon in the opposite hemisphere.

radiometric dates in the abstract. Similarly Londry and Symons (1976) report that Keewatin volcanics, Temiskaming sediments, and Haileyburian diorites in the Noranda area have a metamorphic remanence (pole 4) dated at ≈2500 Ma (Dallmeyer *et al.* 1975). At this time the Algoman granites (pole 5) acquired a primary remanence; however, sampling and radiometric details are not given in the abstract.

The Otto stock in the Superior province gives radiometric ages ranging from 1730 to 2520 Ma, by several methods (Cavanaugh and Seyfert 1977) and it has been assigned an age of ≈2450 Ma. Pullaiah and Irving (1975) (pole 6) drilled 10 of 18 accepted sites in lamprophyre dikes that cut, and may be much younger than, the stock. Also 8 of the 10 lamprophyre sites come from one box-work dike representing a single magmatic pulse. Cavanaugh and Seyfert (1977) (pole 7) do not give any sampling details and their α_{95} of 19° is rather large.

The lower Huronian Thessalon volcanics (pole 8) of this study have an assigned age of ≈2375 Ma. The Mugford basalts of the Nain province yield a similar pole (pole 9; Murthy and Deutsch (1972)) and age of ≈2375 Ma (Barton 1975). Christie *et al.* (1975) report a pole (pole 10) for unmetamorphosed Kaminak dikes in the Churchill province. This pole is based on only three dikes and has a fairly large α_{95} error limit of 17°. Also the radiometric ages have rather broad limits of 2370 ± 200 Ma.

Two paleomagnetic studies on the mid-Huronian Gowganda Formation (≈2300 Ma) differ. Based on an inconclusive fold test using data from the Elliot Lake area, Symons (1975) contends that the sediments acquired their remanence during burial diagenesis (pole 11). Morris (1977) contends that the sediments acquired a primary depositional remanence (DRM) (pole 12) and that in some areas this became a secondary late Huronian (≈2200 Ma) metamorphic remanence (pole 13) that corresponds to Symons' pole 11 (1975). Problems are evident with Morris' interpretation (1977). The primary pole (12) assumes that the Chibougamau and Gowganda Formation are correlative. Separation of poles 12 and 13 assumes that hematization of detrital magnetite occurred during metamorphism whereas in modern (i.e. Mesozoic–Cenozoic) sediments it occurs during deposition and diagenesis. Except for the Whitefish Falls sites, fold tests are based on only 4–7 sites in gently dipping strata so that pre- and post-folding remanence directions are not statistically different. Finally, the Whitefish Falls sites were metamorphosed to upper greenschist and amphibolite facies after intrusion of the Nipissing diabase and so these sites would not record a relatively mild pre-Nipissing event (Card 1976).

Symons (1967) reports a poorly defined pole (pole 14 with an α_{95} of 25°) for the upper mid-Huronian Firstbrook Member based on a reversed remanence direction. Roy and Lapointe (1976) report a well-defined and discordant Firstbrook pole (pole 15) based on a normal remanence direction. Contact tests with Nipissing diabase suggest both poles are primary. Subsequently Stupavsky *et al.* (1976) in an abstract report that one section of the Firstbrook Member contains three normal and three reversed polarity zones. They find that Symons (1967) sampled at the top of the R3 zone, that Roy and Lapointe (1976) sampled entirely in the uppermost N1 zone, that the normal and reverse zones are not antiparallel, and that the reverse zones give directions close to Roy and Lapointe's post-Nipissing metamorphic east direction (1976).

The Abitibi ENE–W dikes pole (pole 16) of Irving and Naldrett (1977) is based primarily on data from Larochelle (1966). The Abitibi ENE–N and NNE dikes (Larochelle 1966) are represented by

only two and one dike, respectively, and so their poles are omitted. The Abitibi dikes of the Superior province give ages of ≈2200 Ma (Gates and Hurley 1973).

The Big Spruce complex pole (pole 17) is based on the stable high temperature 'D' magnetization that Irving and McGlynn (1976) believe to have been acquired during emplacement ≈2170 Ma ago. Post-emplacement tectonic rotation of this pole is likely because the complex apparently lies in a major fault zone, because high-level rocks are exposed on the south side and low-level rocks on the north side of the complex, and because the host Slave province may have been involved in plate motion as noted earlier.

Larson *et al.* (1973) report poorly grouped directions from three recrystallized dikes in the Beartooth Mountains of Wyoming. Problems in the radiometric dating and tectonic interpretation have led to the omission of these results.

Most paleomagnetic studies of the ≈2160 Ma Nipissing diabase in the Superior and Southern provinces have yielded two pole positions — one from reverse polarizations is near the equator offshore of Peru, and one from normal polarizations is in northern North America. Both poles are supported by positive contact tests. The equatorial pole (pole 18) is considered to be a primary thermoremanent magnetization (TRM) by Symons and Londry (1975), and to be a secondary metamorphic remanence by Roy and Lapointe (1976). Conversely, the northern pole is considered by Symons and Londry (1975) to be a Cenozoic unroofing or viscous remanence, and to be a primary TRM (pole 19) by Roy and Lapointe (1976).

The age of the Otish gabbro (pole 20) (Fahrig and Chown 1973) is speculative. It intrudes the Otish Mountain Group that may be Huronian at ≈2300 Ma, and it is apparently correlative with basic intrusives giving minimum K–Ar ages of ≈1950 Ma. The assigned age of ≈2125 Ma may be rather young. This unit is just north of the Grenville Front although there is no evidence of structural deformation.

The Owl Creek dikes in Wyoming have been dated at ≈2100 Ma; however, the pole (pole 21) (Larson *et al.* 1973) is based on data from only three dikes. The Indin dikes in the Slave province (pole 22) (McGlynn and Irving 1975) give a wide range of radiometric ages with a Rb–Sr isochron by Gates and Hurley (1973) of 2093 ± 86 Ma considered to be the most reliable. The Indian Harbour dikes in the Nain province (pole 23) (Murthy and Deutsch 1972) give a minimum K–Ar age of 2080 ± 42 Ma and were sampled over a rather small area. Finally, the Wind River dikes in Wyoming (pole 24) (Spall 1971) give radiometric ages ranging from 1300 Ma to the accepted 2060 Ma. The pole is based on data from four dikes giving a large α_{95} of 24°.

Given the many limitations of the available pole positions it is inadvisable to construct a complex APW path for the Huronian such as constructed by Morris (1977). The Huronian path shown in Fig. 7 is comparable to those proposed by Symons (1975) and by Cavanaugh and Seyfert (1977). The poles from the Slave province are discordant in this construction. This supports the hypothesis of McGlynn and Irving (1975) and Cavanaugh and Seyfert (1977) that the Slave was a separate plate from the Superior–Southern–Nain–Churchill plate.

Huronian Paleoclimate

Symons (1975) summarizes the geological evidence for the glacial origin of the Gowganda Formation 'tillites'. He finds that this unit was deposited at a polar or glacial paleolatitude of 62°. Roy and Lapointe (1976) and Morris (1977) confirm this result. The Thessalon volcanics give a paleolatitude of 35°. There are two consequences. First, the contemporaneous uraniferous Mississagi pyritic quartz-pebble conglomerates of the Blind River – Elliot Lake mining district were formed in a warm temperate climatic belt. Second, the Huronian Supergroup began deposition in a warm temperate paleolatitude of 35° that increased to a cold polar paleolatitude of 65° by the time the mid-Gowganda was deposited.

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